# SemiPSENet: A Novel Semi-Supervised Change Detection Network for Remote Sensing Images

Lei Hu, *Member, IEEE*, Supeng Li, Jiachen Ruan, and Feng Gao, *Member, IEEE*

***Abstract*— Currently, high-precision fully supervised methods are mainly used in the change detection tasks of high-resolution remote sensing images, which require a large amount of labeled data. However, labeling huge amounts of sample data is costly. In order to solve this limitation, a SemiPSENet based on consistent regularization is proposed. SemiPSENet consists of two parts: a supervised phase and an unsupervised phase, which is trained jointly with a small amount of labeled data and a large amount of unlabeled data. In the supervision phase, to generate a multiscale feature map that can fuse more context information, a GELU_PSP module is designed, which can greatly reduce the loss of semantic information. The squeeze-and-excitation (SE) attention module is added after GELU_PSP module, which can highlight important changing features and neglect irrelevant information based on the importance of each feature channel. In the unsupervised phase, to process the unlabeled data, the consistency regularization method is adopted, which can make the predictive change map have the consistency ability under different random perturbations, so that the semantic information can fully be used in the unlabeled image. Experiments on two public datasets LEVIR-CD and WHU show that the proposed SemiPSENet can achieve excellent detection results by relying on the training of a small amount of labeled data and a large amount of unlabeled data. Notably, our network performs better than other SOTA methods when using the same amount of labeled data.**

***Index Terms*— Change detection, consistent regularization, GELU_PSP module, remote sensing image, semi-supervised learning.**

## I. Introduction

THE purpose of change detection for remote sensing images is to identify semantic changes that occur in the same area obtained at different times. The change detection methods can help us quickly obtain the change information, and this has been widely used in disaster assessment [1], land use mapping [2], urban expansion monitoring [3], and other fields.

At present, the main change detection algorithms used in the change detection task are the full supervision methods based on deep learning. For example, Daudt et al. [4] propose the FC-EF network by improving the UNet network [5], and FC-Siam-conc and FC-Siam-diff networks by further integrating with Siamese network models. In [6], a dual attention fully convolutional Siamese network (DASNet) is proposed, which introduces spatial attention mechanism and channel attention mechanism into the Siamese network to obtain more discriminative feature representations, thereby generating more accurate change maps. In [7], a Siamese network SNUNet-CD based on dense connections is proposed, which presents an ensemble channel attention module (ECAM) that can extract the most representative features in the image. In [8], a spatiotemporal enhancement and interlevel fusion network (SEIFNet) is proposed to improve the ability of feature representation for changing objects. A spatiotemporal difference enhancement module (ST-DEM) is introduced to obtain global and local information from the two-time feature maps of each level, and an adaptive context fusion module (ACFM) is proposed to integrate interlayer features under the guidance of different semantic information to form a progressive decoder.

Usually, these full supervision methods require a large amount of labeled data to achieve high detection accuracy. However, the high cost of making labels for massive unlabeled high-resolution remote sensing images greatly limits the application and promotion of fully supervised methods in change detection tasks.

To address this limitation, the researchers have proposed three different solutions. The first method is unsupervised learning [9], although there is no need for labels on training data, the actual application mainly focuses on clustering [10], clustering high similarity data together, which does not perform well in most complex scenarios. The second method is weakly supervised learning [11], which helps generate pseudo labels by using coarse-grained labels, thus guiding model supervised learning. However, compared to fully supervised methods, although the weakly supervised learning methods can save the cost of making labels to some extent, the experimental detection accuracy obtained is not ideal enough, so it does not meet the actual needs. The third method is semi-supervised learning [12], which is a combination of supervised learning and unsupervised learning. It uses a small amount of labeled data and a large amount of unlabeled data for training and can achieve some good detection results. Moreover, an ideal change detection model should be able to detect all changes in the specific areas, while also avoiding complex irrelevant

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changes commonly seen in remote sensing images, such as false changes caused by dark cloud coverage, building shadows, and different sensing angles.

In recent years, with the continuous breakthrough of deep learning technology in the field of change detection on remote sensing images, researchers have proposed many semi-supervised learning methods. For example, with some labeled data, Bovolo et al. [13] propose a semi-supervised change detection method for multispectral images, by SVM technology and an unsupervised model selection strategy based on similarity measurement. Chen et al. [14] propose a semi-supervised change detection method using Gaussian process (GP), which utilizes labeled data and unlabeled data through probabilistic GP classifier and then utilizes context information through MRF regularization. In addition, Yuan et al. [15] propose a semi-supervised change detection method using metric learning. By learning an appropriate distance metric, the unchanged class pixels are closely mapped, while the class pixels about change are mapped as far as possible, and then the unlabeled data are incorporated into the change detection model through a Laplacian regularization framework. Peng et al. [16] propose a semi-supervised change detection method based on generative adversarial networks (GANs) model. This method is composed of two independent neural networks, generator and classifier, and the generator with higher quality and the classifier with strong judgment ability are obtained by repeatedly training on labeled data and unlabeled data. Mittal et al. [17] propose the S4GAN model, which is based on GAN to predict the difference between the segmentation graph and the true segmentation graph by the minimization of the feature matching loss while using the self-training loss to balance the generator and discriminator. Bandara and Pate [18] propose a semi-supervised change detection model with revisiting consistency regularization, which we named SemiRCRNet. SemiRCRNet adopts an encoder and decoder structure while using a small amount of labeled data and a large amount of unlabeled data for the consistency training. Combining unsupervised consistency loss and supervised cross entropy (CE) loss [19], SemiRCRNet network applies small random disturbances to the deep feature difference map and makes the change map obtained by the network consistent with the change map without any disturbance. In [18], it is proved by experiments that the clustering hypothesis of change detection exists in the hidden feature difference domain, and semi-supervised change detection can also benefit from the consistency regularization method. So change detection can benefit from freely obtained unlabeled remote sensing images. Moreover, among the many semi-supervised learning methods, consistency training has shown state-of-the-art performance in different applications, such as classification [20], [21], [22], semantic segmentation [17], [23], [24], and object detection [25].

Inspired by the SemiRCRNet method, in this article, we propose a new semi-supervised change detection network SemiPSENet. The model is divided into two parts: a supervised phase and an unsupervised phase. The model adopts encoding–decoding structure. In order to better generate different maps, we add a Siamese network to extract the feature information of the bi-temporal images separately. In the supervision phase, we use the GELU_PSP module to generate multiscale feature maps with better integrated context information, which can reduce the loss of semantic information. In the supervision phase, we further use the squeeze-and-excitation (SE) [26] attention module to extract important change features and ignore irrelevant change information. In the unsupervised phase, the GELU_PSP module is also used to generate multiscale feature maps integrating context information, and then small random jitter is applied to the deep feature difference map, and the change map obtained by the network is consistent with the change map without any disturbance by using the consistency regularization method. In this article, our SemiPSENet can effectively use the information from labeled and unlabeled multitemporal remote sensing images, thereby improving the performance of change detection.

The main contributions of this article are given as follows.

1) We propose a semi-supervised change detection network based on consistency regularization, which can effectively utilize the semantic information of a small number of labeled remote sensing images and lots of unlabeled remote sensing images, thereby improving the performance of change detection.
2) Gaussian error linear units (GELUs) are used to improve the representation ability of the model. Our proposed GELU_PSP module can generate multiscale feature maps integrating more context information, which can reduce the loss of semantic information and avoid some missing detection.
3) The SE module is introduced in the supervised phase. By learning the relationship between different channels, the network can pay attention to more important change features and filter out irrelevant change information, which helps to identify pseudo-changes and thus improve the detection accuracy of the model.
4) The experimental results on the LEVIR-CD and WHU datasets show that our model can significantly improve change detection results in the case of a small number of label data and it is superior to typical advanced fully supervised or semi-supervised change detection methods.

Organizational structure of the article: In the remaining chapters of this article, Chapter 2 introduces the theoretical knowledge and modules involved in the relevant work. In Chapter 3, the network is proposed in this article, as well as the core modules and loss functions of network are introduced. In Chapter 4, comparative experiments and ablation experiments are conducted between the proposed method and typical advanced methods. Finally, a summary of the work done in this article is provided.

## II. Related Work

In this section, we will briefly describe consistent regularization methods and semi-supervised learning methods, as well as the attention module.

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### A. Consistency Regularization

The consistency regularization method is inspired by two interrelated assumptions: the clustering hypothesis and the low-density separation hypothesis. The clustering hypothesis indicates that if two samples belong to the same cluster in the input distribution, they are likely to belong to the same class. The clustering hypothesis indirectly infers the low-density assumption that the decision boundary separating the two classes should be located in the low-density region [27]. Therefore, the consistency regularization method encourages deep networks to randomly perturb unlabeled data points to generate invariant predictions, ensuring that the output of the model should be consistent before and after the input or the model is perturbed. Based on the great advantages of the method of consistency regularization, Laine and Aila [28] propose the II-model, which integrates the idea of consistency regularization and random enhancement techniques to narrow the differences between these predicted images through training, thus improving the robustness of the model.

### B. Semi-Supervised Learning

Semi-supervised learning [12] is a machine learning technology between supervised learning and unsupervised learning. Its main goal is to overcome the shortcomings of supervised learning and unsupervised learning, allowing network models to automatically utilize unlabeled samples without relying on external interaction. Due to the significant challenge of obtaining large-scale labeled datasets for remote sensing images, semi-supervised learning methods are increasingly gaining interest in tasks such as classification, dimensionality reduction, and change detection. For instance, to address the scarcity of training data in hyperspectral image classification, Hong et al. [29] propose a semi-supervised cross-modality learning framework, which leverages a substantial amount of multispectral data of poor discernment to enhance classification performance. Sun et al. [30] propose a semi-supervised change detection framework for remote sensing image based on pseudo-labeling and consistency regularization.

Moreover, in order to build a better network model using unlabeled data, the pseudo label method is proposed, which involves both labeled data and unlabeled data in the training of the model. For unlabeled data, each unlabeled data is assigned a label with the maximum prediction probability during each weight updating and then the labeled unlabeled data is put into the labeled dataset for model training. However, this method is cumbersome and requires manual intervention, and the detection effect is not ideal. At present, in the task of change detection, the deep semi-supervised learning based on the consistency regularization method has greatly improved the detection accuracy due to the use of data enhancement technology. Inspired by the above methods, we combine the consistent regularization method with semi-supervised learning and apply them to the remote sensing image change detection task.

### C. Attention Mechanism

The attention mechanism enables the network to integrate different features more efficiently, which has a significant impact on improving the accuracy of image processing problems, and it has been widely used in computer vision tasks.

Convolutional block attention module (CBAM) [31] applies channel attention and spatial attention to adaptive feature refinement. Ding et al. [32] propose a patch attention module to highlight the focus area of feature maps. Su et al. [33] mainly focus on similar objects in a small batch of images and encode the interaction information between them through a self-attention mechanism. The graph attention (GAT) mechanism [34] fully considers the relationships among input data, enabling it to more accurately capture the correlations among data when processing graph-structured data. SE attention [26] adjusts the feature values of each channel, emphasizes the information of important channels, and suppresses the information of unimportant channels. We focus on the relationship between channels and use the SE attention module to learn the importance of each channel, enhancing the expression ability of the network in the feature representation, thus improving the performance of the model.

## III. Proposed Method

In this section, we will explain the proposed SemiPSENet architecture in detail and then introduce GELU, GELU_ PSP module and SE attention module. Finally, we will introduce the loss functions in the supervised and unsupervised phases as well as the total loss function.

### A. Semi-Supervised Network Architecture

The SemiPSENet network is an improvement network based on the SemiRCRNet [18]. Our network architecture consists of two parts: the supervised phase network PSENet and the unsupervised phase network. The specific structure is shown in Fig. 1. In the supervision phase, the labeled bi-temporal remote sensing image is input to the encoder to obtain a difference map. The GELU_PSP module is used to generate the multiscale feature map with more contextual information, the SE attention module is used to filter out the unimportant information and highlight the important changing features. And, the CE loss is used to calculate the supervision loss $L_{sup}$ of the predicted change map $y^{\wedge l}$ and the real change map $y_{gt}^{l}$. In the unsupervised phase, the unlabeled data as a part of input of the model is processed by encode module and GELU_PSP module to be divided into two branches. Among them, the output, prediction map $y_p^{\sim ul}$, from the red branch is regarded as the benchmark, and the multiple prediction maps by random jitter processing are the outputs of the blue branch. Each output prediction map is compared to the baseline $y_p^{\sim ul}$ by calculating unsupervised loss $\mathcal{L}_{unsup}$ using mean square error. The effect of continuous optimization of the model can be achieved through backpropagation.

*1) Supervised Phase:* This phase uses labeled data. The encoder $f_e$ is used to obtain the hidden features, $F_A^l$ and $F_B^l$. $F_A^l$ corresponds to the image before change $I_A^l$, $F_B^l$ corresponds to the image after change $I_B^l$. $F_A^l$ and $F_B^l$ are defined as follows:

$$F_A^l = f_e\left(I_A^l\right) \tag{1}$$

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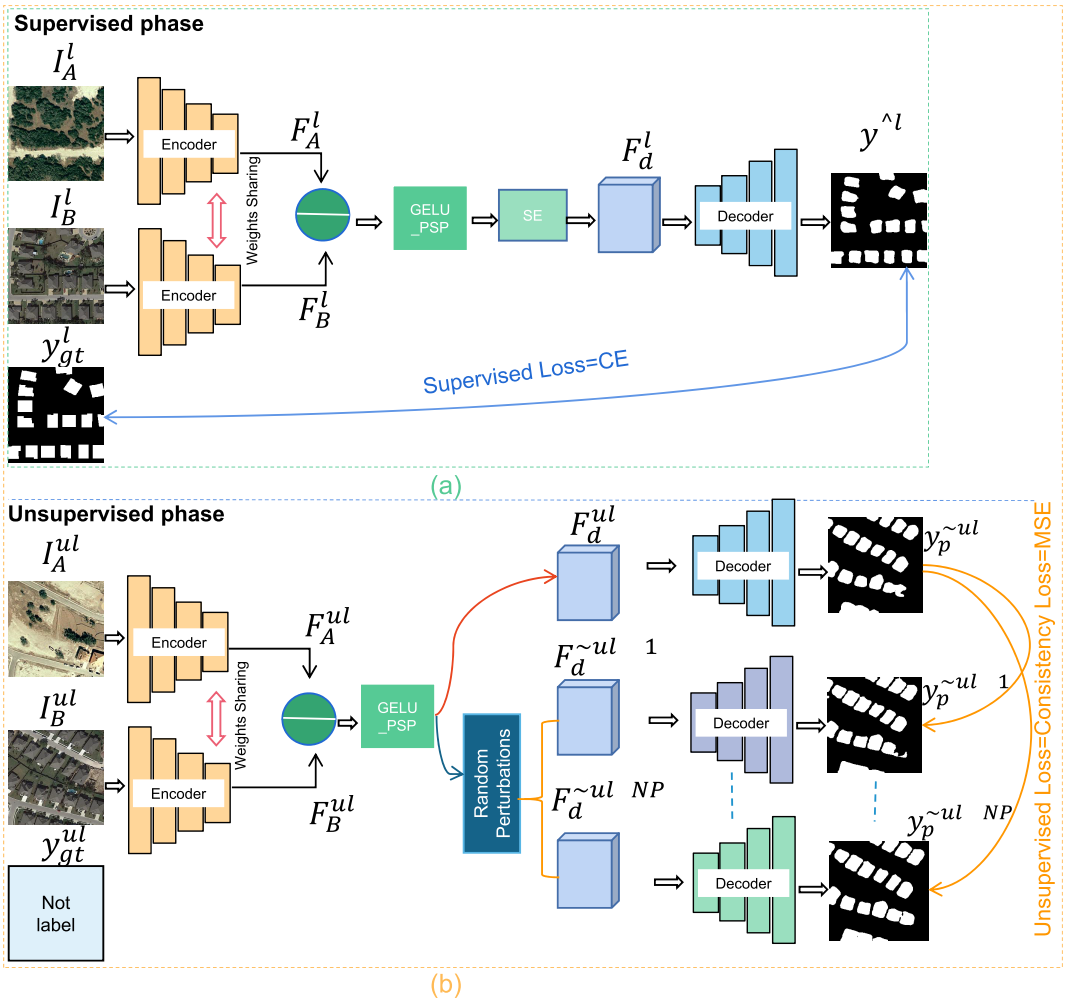



Fig. 1. SemiPSENet network structure. (a) PSENet. (b) SemiPSENet.

$$F_B^l = f_e\left(I_B^l\right). \tag{2}$$

The GELU_PSP module is used to generate multiscale feature maps that incorporate more contextual information, which further reduces missing detection of changing regions. Then, the SE module is used to highlight the important change features and suppress the irrelevant false change information in the image, so as to generate more accurate feature difference map $F_d^l$. The feature difference map $F_d^l$ is formally defined as follows:

$$F_d^l = f_{\text{GELU_PSP}}\left(|F_A^l - F_B^l|_1\right). \tag{3}$$

Then, we use the decoder $f_d$, including a series of subpixel convolutional upsampling blocks, to achieve a prediction map $y^{\wedge l}$ with the same spatial resolution as the input bi-temporal image. The internal process of the decoder can be represented as

$$y^{\wedge l} = f_d\left(F_d^l\right). \tag{4}$$

*2) Unsupervised Phase:* This phase uses unlabeled data. The first half of the processing is the same as that of the supervision stage, where the GELU_PSP module is also used to generate multiscale feature maps integrating context information and retain more semantic information. As shown in Fig. 1, the second half of the processing is different from the supervision phase, this stage generates two different branches after processing by the GELU_PSP module. Except for not using the SE module, the red branch is basically consistent with the processing mode of the supervision stage. And the output of the red branch is $y_p^{\sim \text{ul}}$.

The blue branch in this unsupervised phase is some different, random perturbations processing is performed on the input feature map, so as to generate multiple feature maps with subtle differences, and then the multiple predictive change maps ($y_p^{\sim \text{ul1}}, y_p^{\sim \text{ul2}}, y_p^{\sim \text{ul3}} \ldots y_p^{\sim \text{ulNP}}$) are obtained through the corresponding decoders. Here we use a consistent regularization method to improve model performance. On the one hand, a consistent regularization method can suppress the influence of noise and uncertainty and can improve the robustness of learning algorithm by encouraging the model to generate consistent output or learn stable representation. On the other hand, by encouraging models to produce similar outputs for similar samples, a consistent regularization method can leverage structural information in unlabeled data.

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In the unsupervised stage, the change map outputted by the red branch is very close to the change map outputted by the PSENet network of the supervised stage, so it is used as the baseline $y_p^{\sim ul}$ to calculate the unsupervised loss. Each prediction map $y_p^{\sim ul\,i}(i = 1, \ldots, NP)$ is compared with the baseline $y_p^{\sim ul}$. By calculating unsupervised losses, we try to obtain the optimal weight parameters, with the aim of synthesizing the results of prediction maps to obtain a relatively balanced change map.

*3) Correlation Between Supervised and Unsupervised Phases:* Although the supervised learning phase and unsupervised learning phase are different, they can complement each other. Supervised learning can provide valuable auxiliary information for unsupervised learning, including preprocessing data, providing feature representations, and predicting labels.

After learning the labeled data, supervised learning can provide good feature representations for unsupervised learning and help unsupervised learning algorithms better understand the structure and patterns of data. In our network, during the processing of $y_p^{\sim ul}$ in the unsupervised phase, the encoder and decoder benefit from the supervised phase. During the verification in training our semi-supervised learning algorithm, the model of supervised learning phase can be used to output the predicted label for an unlabeled data, the unsupervised learning algorithm will judge its predicted label by comparing it with the predicted label. The supervised learning phase provides more context and guidance for the unsupervised learning tasks.

## B. Gaussian Error Linear Units

In order to better solve the problem of insufficiency on model expression ability, we introduce the GELU [35]. In [35], experiments have shown that GELU performs better than ReLU [36] and ELU [37] in computer vision, natural language processing, and speech tasks. During the training process of the model, the main function of the activation function is to add nonlinear factors to solve the defect of insufficient ability on model expression, which plays a crucial role in the entire neural network. In the modeling process of neural networks, a very important property of the model is nonlinearity. At the same time, in order to enhance the generalization ability on model, it is necessary to add random regularization. Random regularization and nonlinear activation are two separate things, but the input of the model is determined by both nonlinear activation and random regularization. The GELU introduces the idea of random regularization, which is a high-performance neural network activation function. The GELU activation function is defined as $x\Phi(x)$, where $\Phi(x)$ is the standard Gaussian cumulative distribution function, and $x$ is a linear input.

The formula for the GELU [35]

$$\begin{aligned} \mathrm{GELU}(x) &= xP(X \le x) = x\Phi(x) \\ &= x \cdot \frac{1}{2}\left[1 + \mathrm{erf}\left(x/\sqrt{2}\right)\right]. \end{aligned} \tag{5}$$

On the horizontal interval $[-4, 3]$ and the vertical interval $[-1, 3]$, let the hyper-parameters $\alpha = 1$ in the ELU activation function, and we visually display the three activation functions, GELU, ReLU, and EUL, in the form of images, as shown in Fig. 2.

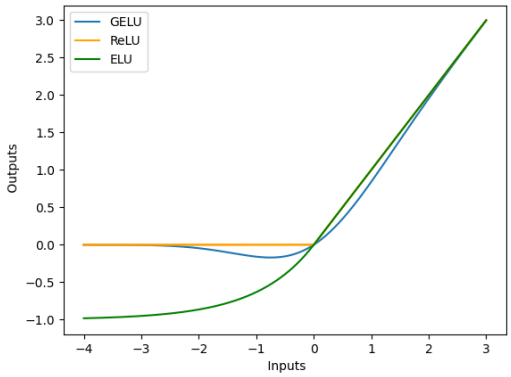



Fig. 2. Three activation function curves.

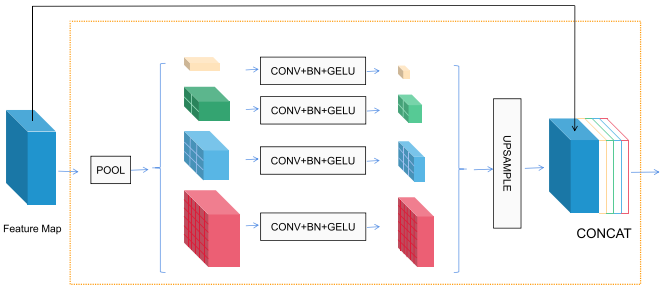



Fig. 3. GELU_PSP module structure.

When the input value is negative, GELU maps the input to a negative value, which helps the network push the weight and bias changes in the right direction, thus avoiding the gradient disappearance problem of the ReLU activation function, which makes the GELU activation function converge faster than the ReLU activation function. Since neural networks do not learn hyper-parameter $\alpha$, the use of ELU is limited by many conditions, and it is difficult to obtain satisfactory results.

## C. GELU_PSP Module

In order to reduce the missing detection of the changed area to a greater extent, we propose the GELU_PSP module, the module structure is shown in Fig. 3. GELU_PSP module integrates the advantages of pyramid pool module (PSP) [38] and GELU, which can be used to generate multiscale feature maps with more context information to the maximum extent, so as to retain more semantic information. Moreover, the fusion of context information and multiscale is also very effective for improving model accuracy, greatly reducing the occurrence of missed detections.

Introduction to the main principle: The pooling of four different scales of kernel size of $1 \times 1$, $2 \times 2$, $3 \times 3$, and $6 \times 6$ is used on the feature map. The convolutional layer is used for feature extraction and BN layer normalization. The GELU is introduced to obtain four different scales of feature maps that fuse more context information.

## D. SE Attention Module

The structure diagram of the SE attention module [26] is shown in Fig. 4. During the Squeeze stage, input features are

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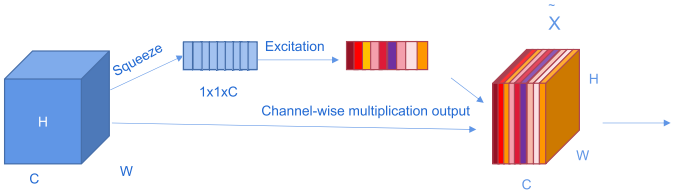



Fig. 4. Structure diagram of the SE attention module.

compressed through global average pooling operation to capture global feature information. In the excitation stage, a pair of fully connected layers and activation functions are used to perform nonlinear mapping and adjustment on compressed features to learn the weight relationship between features. During the model training process, the SE attention module can adaptively select and emphasize important features, improving the discriminative ability of features. In addition, the SE module is a plug-and-play attention module that highlights the features of changing regions in bi-temporal images with minimal computational complexity, while filtering out pseudo-changing regions, resulting in the generation of more accurate change maps.

In our whole network structure, we select SE attention module based on its characteristics. And we have also tried other attention mechanisms, like CBAM, coordinate attention (CA), channel attention module (CAM), and SE attention module performs the best.

### E. Loss Functions

*1) Supervision Loss:* In the supervision phase, we define labeled images as $D_l = \{\{I^l_{A,i}, I^l_{B,i}\}, y^l_i\}_{i=1}^{N_l}$, where $\{I^l_{A,i}, I^l_{B,i}\}$ represents the $i$th bi-temporal image, $y^{\wedge l}_i$ is the corresponding ground-truth change mask, and $N_l$ is the quantity of the labeled dataset. For this purpose, we use CE loss [19] as the supervised loss $L_{\text{sup}}$, calculating the loss of the predicted change map $y^{\wedge l}_i = f_{\text{CD}}(I^l_{A,i}, I^l_{B,i})$ and actual change map $y^l_{\text{gt},i}$, which can be expressed mathematically as

$$L_{\text{sup}} = CE\left(y^{\wedge l}_i, y^l_{\text{gt},i}\right). \tag{6}$$

*2) Unsupervised Loss:* In the unsupervised stage, $\mathcal{D}_{\text{ul}} = \{I^{\text{ul}}_{A,i}, I^{\text{ul}}_{B,i}\}_{i=1}^{N_{\text{ul}}}$ is used to express the unlabeled bi-temporal image, where $\{I^{\text{ul}}_{A,i}, I^{\text{ul}}_{B,i}\}$ is the $i$th unlabeled bi-temporal images, $N_{\text{ul}}$ is the quantity of the unlabeled dataset, it is usually assumed to be higher than the quantity of the labeled dataset. Our goal is to efficiently utilize any raw bi-temporal images to improve the performance of our semi-supervised change detection model $f_{\text{CD}}(\cdot)$. We adopt an unsupervised loss $\mathcal{L}_{\text{unsup}}$ based on unlabeled data, which provides an additional training signal to optimize the parameters of $f_{\text{CD}}(\cdot)$. The feature difference map $F^{\text{ul}}_{d,i}$ obtained by using unlabeled bi-temporal images $\{I^{\text{ul}}_{A,i}, I^{\text{ul}}_{B,i}\}$, while $\{(F^{\sim\text{ul}}_{d,i})_1, \ldots, (F^{\sim\text{ul}}_{d,i})_p, \ldots, (F^{\sim\text{ul}}_{d,i})_{p=N_p}\}$ serves as a set of randomly perturbed versions of hidden feature differences, and the prediction map $y^{\sim\text{ul}}_i$ is obtained through the main decoder. It can be represented by formula as

$$y^{\sim\text{ul}}_i = f_d\left(F^{\sim\text{ul}}_{d,i}\right). \tag{7}$$

Similarly, we process each perturbed version of the hidden feature difference map by using a set of NP auxiliary decoders which are similar in design as $f_d(\cdot)$ and obtain the corresponding prediction maps $(y^{\wedge\text{ul}}_i)_p$. It can be represented by formula as

$$\left(y^{\sim\text{ul}}_i\right)_p = f^p_d\left(\left(F^{\sim\text{ul}}_{d,i}\right)_p\right) p = 1, \ldots, \text{NP}. \tag{8}$$

Unsupervised loss is represented by formula as

$$\mathcal{L}_{\text{unsup}} = \sum_{p=1}^{N_p} d\left(\left(y^{\sim\text{ul}}_i\right)_p, y^{\wedge\text{ul}}_i\right) \tag{9}$$

where d$(\cdot)$ is a distance metric that measures the difference between predictions $(y^{\wedge\text{ul}}_i)_p$, and here $d(\cdot)$ uses the mean square error.

*3) Total Loss:* We define the final loss function $\mathcal{L}$ as follows:

$$\mathcal{L} = L_{\text{sup}} + \lambda(t)\mathcal{L}_{\text{unsup}} \tag{10}$$

where $\lambda(t)$ is a regularization constant that is a function of training about the number of iterations $t$. Specifically, we change $\lambda(t)$ from 0 to 1 just like the Gaussian function in the first iteration $T$ times, and keep it as a constant of 1 when the remaining iterations $t > T$.

## IV. Experiments

In this section, first, we will introduce two typical and public datasets on change detection of remote sensing image. Then, a brief description of the training details, the advanced methods used for comparison, and evaluation metrics will be provided. Next, we will introduce the experimental results of the comparison method to verify the effectiveness of the proposed network. Finally, we will further analyze the ablation experiments to evaluate the performance of our method.

### A. Data Description

*1) LEVIR-CD Dataset:* The LEVIR-CD is a large remote sensing building change detection dataset. The dataset focuses on changes in land use, primarily in buildings covering various types of buildings, such as family villas, high-rise apartments, small garages, and large warehouses. The images have many pseudo changes caused by seasons and lighting, making them a very challenging dataset. We divide the LEVIR-CD dataset into three parts: training set, validation set, and test set. The training set consists of 7120 images, the validation set consists of 1024 images and the test set consists of 2048 images.

*2) WHU Dataset:* The WHU is a dataset primarily used for detecting building changes. Due to the large change area with extremely unbalanced distribution, we can use this dataset to further test the effectiveness of the proposed method. We divided the WHU dataset into 5947 images in the training set, 743 images in the verification set, and 744 images in the test set.

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### B. Implementation Details

*1) Training Settings:* In this article, we build our network model using the Pytorch framework and train it using the NVIDIA RTX 3090 GPU. The learning rate is initially set to 0.01, using the SGD optimizer, with a weight decay of $1 \times 10^{-4}$. We train 80 epochs on the model and use a batch size of 8 for both the labeled and unlabeled datasets. In order to prove the effectiveness of our method, the four different ratios of sampling rate are {5%, 10%, 20%, 40%} in the labeled training data sample [18] and the remaining images in the training set are regarded the unlabeled data by discarding the corresponding change labels. So that we can compare the impact of the rate of the labeled data on the model results. During training, we perform data enhancement through random flipping, random scaling (0.8–1.2), random cropping, Gaussian blur, and random color jitter.

*2) Comparative Networks:* We conduct a lot of comparison and analysis with some advanced change detection methods, including supervised methods and semi-supervised methods. These methods are introduced as follows.

1) *FC-EF [4]:* It is a typical supervised method, and its fully convolutional network extracts and analyses image feature information through the UNet structure, which concatenates bi-temporal image channels as the input of the model.
2) *FC-Siam Conc [4]:* It is a supervised method. It adds a Siamese structure to the FC-EF network, allowing weights to be shared. During skip connections, the corresponding layer of the decoder and the feature maps from two branch networks are overlaid on the channel dimension.
3) *PSENet:* It is a method based on the part of supervision phase of SemiRCRNet [18], which adds a GELU_PSP module between the encoder and decoder to extract global information of feature maps and then uses the SE module to highlight important changing features.
4) *S4GAN [17]:* It is a semi-supervised method that uses unlabeled data and predicts the difference between the change map and the real change map by minimizing feature matching loss.
5) *SemiCDNet [16]:* It is a semi-supervised method using GAN. By using generators and discriminators to improve the consistency of feature distribution and entropy maps between labeled and unlabeled data, only labeled data are used and unlabeled data are not utilized during model training.
6) *SemiRCRNet [18]:* It is a semi-supervised model trained using both labeled and unlabeled bi-temporal images. In this model, supervised cross-entropy loss and unsupervised change loss are formulated.
7) *FCD-GAN [39]:* It is a framework named fully convolutional change detection with GAN, which can conclude unsupervised, weakly supervised, regional supervised, and fully supervised change detection into one framework. FCD-GAN presents a pioneering work on the task of weakly supervised change detection. It introduces a novel approach for weakly supervised segmentation using GANs.
8) *SemiSANet [30]:* It is a semi-supervised framework for remote sensing image change detection based on pseudo-labeling and consistency regularization, by making the change detection results of distorted images consistent with the pseudo-label, the performance and robustness of the model are improved.

For the supervised method, the rate of the training set in the total sample is the labeled data in the semi-supervised method, and the remaining data is used as the test set.

*3) Evaluation Metrics:* In order to effectively evaluate the effectiveness of different methods, we use overall accuracy (OA), $F1$-score, Kappa coefficient (Kappa), and intersection over union (IOU) as evaluation metrics. They are defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{11}$$

$$\mathrm{IOU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{12}$$

$$F1 = \frac{2 \times P \times R}{\mathrm{P} + \mathrm{R}} \tag{13}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{14}$$

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{15}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{16}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FN}) \times (\mathrm{TP} + \mathrm{FP}) + (\mathrm{FN} + \mathrm{FP}) \times (\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} \tag{17}$$

where TP represents the number of true positives, FP represents the number of false positives, TN represents the number of true negatives, and FN represents the number of false negatives.

### C. Results

We summarize and analyze the experimental results on the two datasets.

*1) LEVIR-CD Dataset:* We selected small targets, medium targets, complex targets, and large targets as four types of changing targets. In order to analyze these models more conveniently and visually, we select prediction change maps with a sampling rate of 20% from the experimental results for analysis, as shown in Fig. 5. It can be seen that the change map [Fig. 5(l)] obtained by the proposed SemiPSENet is basically the same as the ground truth [Fig. 5(c)]. Our method outperforms other comparison methods in terms of boundary accuracy, missed detections, and false detections.

The proposed SemiPSENet can overcome missed detection about similar color. When there exists a change between building and surrounding, most other methods judge it as unchanged area, but our SemiPSENet makes the right judgment with finer contours. Our method uses a small number of labeled data and a large amount of unlabeled data for joint training, and experiments show that this can improve the detection performance of the model.

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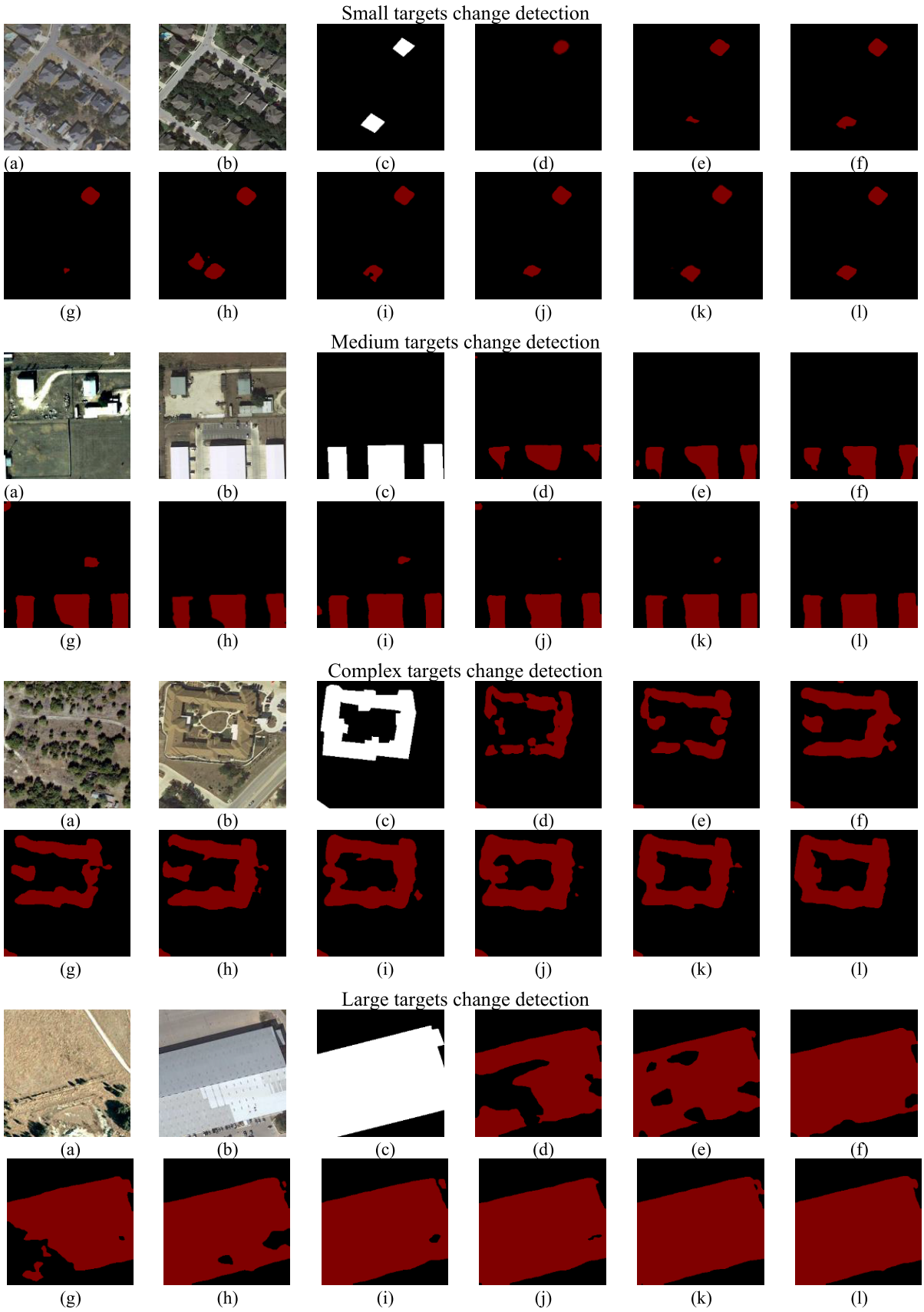



Fig. 5. Comparison on LEVIR-CD dataset using 20% labeled images for training. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-Conc. (f) PSENet. (g) S4GAN. (h) SemiCDNet. (i) SemiRCRNet. (j) FCD-GAN. (k) SemiSANet. (l) SemiPSENet.

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TABLE I
RESULTS OBTAINED BY DIFFERENT METHODS ON THE LEVIR-CD DATASET

| Methods | Labeled Sampling Ratio | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 5% | | | | 10% | | | | 20% | | | | 40% | | | |
| | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa |
| **FC-EF** | 0.5667 | 0.9672 | 0.2963 | 0.2556 | 0.6313 | 0.9804 | 0.4552 | 0.4216 | 0.7036 | 0.9832 | 0.6379 | 0.6155 | 0.7316 | 0.9835 | 0.6512 | 0.6276 |
| **FC-Siam-Conc** | 0.5846 | 0.9746 | 0.5415 | 0.5821 | 0.6656 | 0.9813 | 0.6312 | 0.6483 | 0.7256 | 0.9854 | 0.7252 | 0.7154 | 0.7428 | 0.9864 | 0.7915 | 0.7853 |
| **PSENet** | 0.6338 | 0.9790 | 0.6054 | 0.5896 | 0.6972 | 0.9830 | 0.7152 | 0.7083 | 0.7309 | 0.9851 | 0.7761 | 0.7632 | 0.7439 | 0.9860 | 0.8235 | 0.8298 |
| **S4GAN** | 0.6318 | 0.9780 | 0.5126 | 0.5068 | 0.6683 | 0.9821 | 0.7428 | 0.7395 | 0.7340 | 0.9861 | 0.7984 | 0.7842 | 0.7522 | 0.9862 | 0.8474 | 0.8395 |
| **SemiCDNet** | 0.6623 | 0.9817 | 0.6524 | 0.6497 | 0.7146 | 0.9843 | 0.8056 | 0.7983 | 0.7421 | 0.9860 | 0.8256 | 0.8198 | 0.7535 | 0.9864 | 0.8357 | 0.8298 |
| **SemiRCRNet** | 0.725 | 0.9847 | 0.7782 | 0.7655 | 0.755 | 0.9863 | 0.8427 | 0. 8378 | 0.762 | 0.9865 | 0.8584 | 0.8511 | 0.772 | 0.9872 | 0.8650 | 0.8604 |
| **FCD_GAN** | 0.6574 | 0.9849 | 0.7883 | 0.7795 | 0.7217 | 0.9874 | 0.8225 | 0.8194 | 0.7481 | 0.9855 | 0.8412 | 0.8345 | 0.7632 | 0.9863 | 0.8524 | 0.8465 |
| **SemiSANet** | 0.6800 | 0.9852 | 0.8095 | 0.8019 | 0.7440 | **0.9881** | 0.8532 | 0.8470 | 0.7698 | **0.9895** | 0.8699 | 0.8644 | - | - | - | - |
| **SemiPSENet** | **0.7515** | **0.9865** | **0.8512** | **0.8435** | **0.7740** | 0.9876 | **0.8674** | **0.8547** | **0.7819** | 0.9880 | **0.8894** | **0.8765** | **0.7905** | **0.9884** | **0.9132** | **0.9045** |

We summarize the values of IOU, OA, F1, and Kappa about SemiPSENet and the comparison methods in Table I. When the sampling rate is 5%, the overall performance of the supervised methods of FC-EF and FC-Siam-Conc is significantly weaker than other semi-supervised methods, especially on the metrics of F1 and Kappa. With the increase of sampling rate, the detection performances of each method improve, which strongly demonstrates that using more labeled data can indeed improve model performances. Compared with the supervised methods, S4GAN and SemiCDNet adopt GAN to generate pseudo labels from unlabeled data and then use the hidden information in the pseudo labels to assist in training the network. Compared with the S4GAN and SemiCDNet methods, SemiRCRNet utilizes labeled and unlabeled data for joint training. With a sampling rate of only 5%, its IOU and OA values can reach as high as 0.725 and 0.9847, surpassing the performance metrics of FC-EF, FC-Siam Conc, PSENet, S4GAN, and SemiCDNet networks that use a sampling rate of 10%. This greatly solves the problem of model performance degradation caused by the small amount of labeled data.

In Table I, it is worth noting that the SemiPSENet model proposed in this article achieves the best performance among all the comparative SOTA methods. The SemiPSENet achieves a high accuracy at the cost of 5% labeled data, and its IOU value is 0.7515 and OA value is 0.9865, which exceeds the calculation results of the fully supervised change detection model with a sampling rate of 40%. Moreover, compared to the semi-supervised methods S4GAN and SemiCDNet with a sampling rate of 40%, the IOU values of the three methods are closer, but the OA values calculated using the SemiPSENet method are higher. Under the same conditions, compared with the semi-supervised SOTA method SemiRCRNet, the parameter indexes of SemiPSENet are optimal, especially in the IOU index, which improves by about 2%.

When the sampling rate is 10% or 20%, the OA values of SemiPSENet and SemiSANet are very close, and the other three metrics are that SemiPSENet method is better. The performance of the SemiPSENet is enhanced when the number of labeled images increases at all sampling rates. When the sampling rate is 40%, the SemiPSENet can achieve excellent performance with an F1-score of 0.9132, Kappa of 0.9045, OA of 0.9884, and IOU of 0.7905.

*2) WHU Dataset:* We also performed the same experiments on the WHU dataset as the LEVIR-CD dataset.

Some experimental results of change detection maps with a sampling rate of 20% are shown in Fig. 6. We also selected four kinds of test targets, small targets, medium targets, complex targets, and large targets, in the WHU dataset. These changing objects in the WHU dataset have a greater variety of shapes, sizes, and colors, and many of the changed houses do not have a clear 3-D on the map, which makes many model changes difficult to detect.

In Fig. 6, the change maps [Fig. 6(l)] obtained by our method are very similar to the ground truth [Fig. 6(c)]. Moreover, in terms of boundary accuracy, missed detection, and false detection, the change areas obtained by our method are better than most other comparison methods.

In Table II, the experimental results about evaluation metrics on the WHU dataset are summarized. The proposed SemiPSENet also achieves the overall best performance. When the sampling rate is 5%, the IOU and OA values calculated by the SemiPSENet are 0.6788 and 0.9856, respectively. The IOU value even exceeds the fully supervised change detection methods FC-EF, FC-Siam-Conc, and PSENet with a sampling rate of 10%. In particular, the IOU value exceeds the calculation results of the semi-supervised change detection methods SemiRCRNet and SemiSANet by more than 2%. Thus, under the same conditions, the SemiPSENet is better than or close to the semi-supervised SOTA method SemiSANet. As the amount of labeled data increases, the performance of these models improves significantly.

Furthermore, on the metric of F1 and Kappa, the SemiPSENet method is slightly higher than the SOTA method SemiSANet or SemiRCRNet, whether the sampling rate is 5% or 10%, or the sampling rate is 20% or 40%.

The main reasons for our proposed method achieving the outstanding performance are as follows.

First, the GELU_PSP structure is used in both the supervised and unsupervised stages of the network structure, which can obtain multiscale features of integrated contextual

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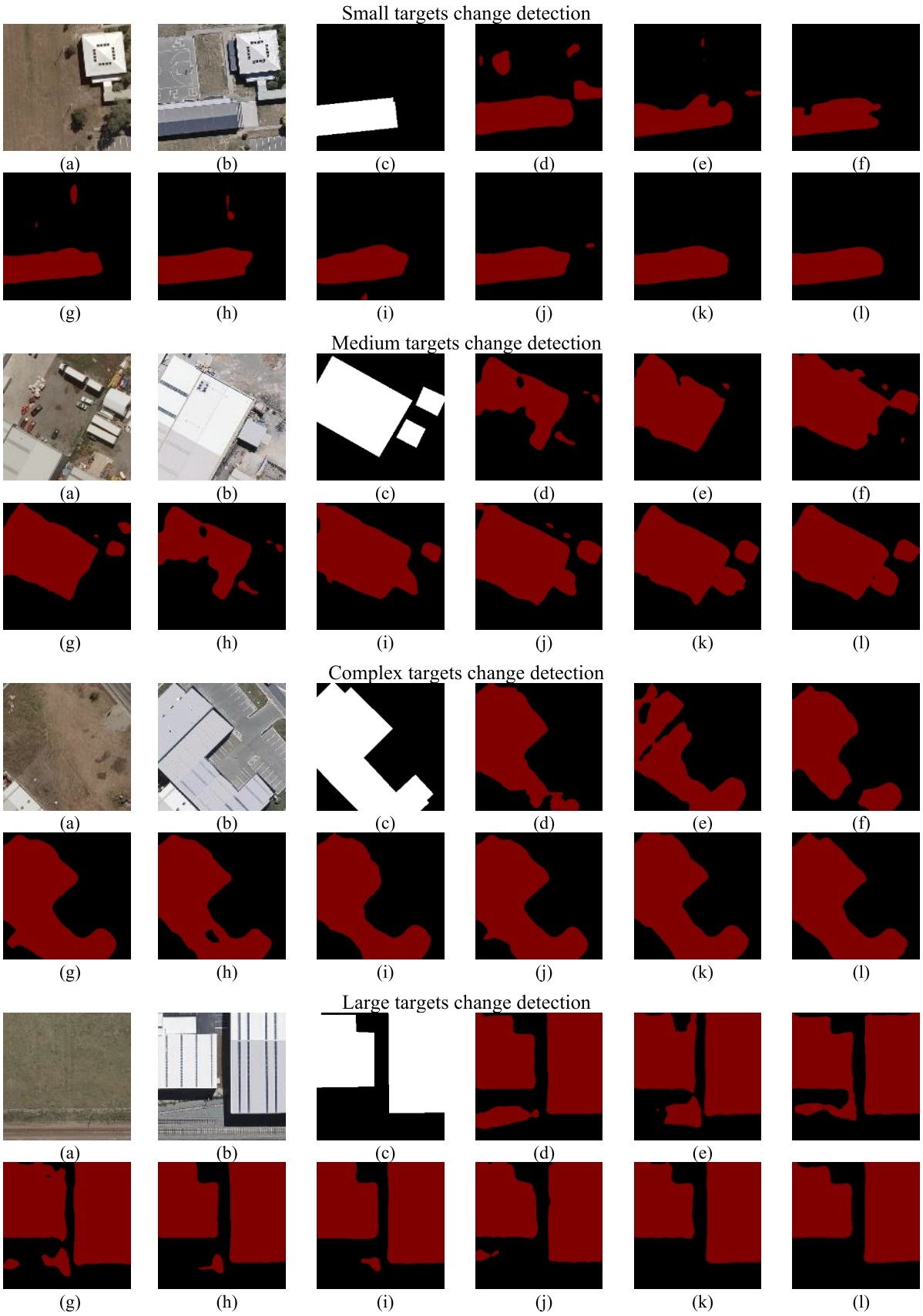



Fig. 6. Comparison on WHU dataset using 20% labeled images for training. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-Conc. (f) PSENet. (g) S4GAN. (h) SemiCDNet. (i) SemiRCRNet. (j) FCD-GAN. (k) SemiSANet. (l) SemiPSENet.

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TABLE II
RESULTS OBTAINED BY DIFFERENT METHODS ON THE WHU DATASET

| Methods | Labeled Sampling Ratio | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 5% | | | | 10% | | | | 20% | | | | 40% | | | |
| | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa |
| **FC-EF** | 0.4767 | 0.9656 | 0.4852 | 0.4277 | 0.5876 | 0.9712 | 0.5284 | 0.4613 | 0.6324 | 0.9805 | 0.5521 | 0.4833 | 0.7349 | 0.9856 | 0.5664 | 0.5087 |
| **FC-Siam-Conc** | 0.4953 | 0.9736 | 0.4054 | 0.3648 | 0.5924 | 0.9803 | 0.4529 | 0.4165 | 0.6518 | 0.9827 | 0.5162 | 0.4888 | 0.7478 | 0.9883 | 0.5432 | 0.5254 |
| **PSENet** | 0.5483 | 0.9809 | 0.5832 | 0.5154 | 0.6006 | 0.9828 | 0.6201 | 0.6121 | 0.6540 | 0.9835 | 0.6558 | 0.6394 | 0.7668 | 0.9897s | 0.6832 | 0.6522 |
| **S4GAN** | 0.4624 | 0.9760 | 0.4665 | 0.3764 | 0.6192 | 0.9822 | 0.4835 | 0.3929 | 0.7071 | 0.9852 | 0.5232 | 0.4363 | 0.7643 | 0.9892 | 0.6354 | 0.5933 |
| **SemiCDNet** | 0.5328 | 0.9790 | 0.6263 | 0.5685 | 0.6113 | 0.9810 | 0.6861 | 0.6352 | 0.6659 | 0.9836 | 0.7124 | 0.6737 | 0.7513 | 0.9888 | 0.7429 | 0.6952 |
| **SemiRCRNet** | 0.658 | 0.9837 | 0.7294 | 0.7842 | 0.681 | 0.9847 | 0.8014 | 0.8252 | 0.748 | 0.9884 | 0.8461 | 0.8258 | 0.772 | 0.9896 | 0.8696 | 0.8335 |
| **FCD_GAN** | 0.6251 | 0.9758 | 0.6926 | 0.7232 | 0.6544 | 0.9765 | 0.7424 | 0.7819 | 0.6865 | 0.9811 | 0.7887 | 0.7652 | 0.7004 | 0.9854 | 0.8229 | 0.8098 |
| **SemiSANet** | 0.6589 | 0.9611 | 0.7729 | 0.8019 | **0.7172** | 0.9693 | 0.8353 | 0.8184 | **0.7834** | 0.9778 | 0.8786 | 0.8664 | - | - | - | - |
| **SemiPSENet** | **0.6788** | **0.9856** | **0.7925** | **0.8225** | 0.7057 | **0.9860** | **0.8561** | **0.8317** | 0.7765 | **0.9897** | **0.8994** | **0.8921** | **0.7861** | **0.9905** | **0.9223** | **0.9175** |

TABLE III
RESULTS OF ABLATION EXPERIMENTS ON THE LEVIR-CD DATASET

| Methods | Module | | Labeled Sampling Ratio | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SE | GELU _PSP | 5% | | | | 10% | | | | 20% | | | | 40% | | | |
| | | | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa |
| **SemiRCRNet** | | | 0.725 | 0.9847 | 0.7782 | 0.7655 | 0.755 | 0.9863 | 0.8427 | 0.8378 | 0.762 | 0.9865 | 0.8584 | 0.8511 | 0.772 | 0.9872 | 0.8650 | 0.8604 |
| **SemiPSENet** | √ | | 0.7421 | 0.9860 | 0.8455 | 0.8384 | 0.7723 | 0.9875 | 0.8574 | 0.8452 | 0.7713 | 0.9870 | 0.8785 | 0.8705 | 0.7831 | 0.9880 | 0.9045 | 0.8998 |
| **SemiPSENet** | | √ | 0.7452 | 0.9861 | 0.8501 | 0.8411 | 0.7685 | 0.9874 | 0.8632 | 0.8468 | 0.7727 | 0.9877 | 0.8755 | 0.8649 | 0.7901 | 0.9884 | 0.9087 | 0.9015 |
| **SemiPSENet** | √ | √ | **0.7515** | **0.9865** | **0.8512** | **0.8435** | **0.7740** | **0.9876** | **0.8674** | **0.8547** | **0.7819** | **0.9880** | **0.8894** | **0.8765** | **0.7905** | **0.9884** | **0.9132** | **0.9045** |

information to a greater extent, reducing the missed detection of changing regions to a certain extent.

Second, this article adopts an encoding–decoding structure. In order to better identify pseudo changes, an SE module is added in the supervision stage of the network structure, so that the network focuses on certain feature channels and suppresses feature channels that are not useful for the current task to highlight important features, which can reduce some false detection.

### *D. Ablation Study*

In order to better determine the impact of each module in the model on the change detection task, we design ablation experiments to explore the impact of the GELU_PSP module and the SE attention module on the performance of the model.

The SemiRCRNet is the baseline method with PSP, and the proposed SemiPSENet adds the GELU_PSP module or the SE attention module.

*1) LEVIR-CD Dataset:* The ablation experimental results of the proposed method on LEVIR-CD are shown in Table III. From Table III, when adding the GELU_PSP module or the SE attention module, it can be analyzed that the OA value increases slightly, whether the sampling rate is 5% or 10%, or the sampling rate is 20% or 40%, due to the OA value is very high. But, the IOU value increases from 0.93% to 2.65%. The F1 value increases from 1.47% to 7.30%. The Kappa value increases from 0.74% to 7.80%.

In order to further visually reflect the influence of each module in the ablation experiment, we show the resulting graphs, as shown in Fig. 7. Among them, three groups of representative graphs with a sampling rate of 40% in LEVIR-CD dataset are selected in Fig. 7. From these graphs, it can be seen that the change maps obtained by the basic network of SemiRCRNet have missed detection of some change regions. When the GELU_PSP module is added between the encoder and the decoder, the generated change map can reduce the missing detection of the change areas; and after the SE attention module is added to the basic network, the visual effect of the change maps can also be improved. When both the SE attention module and the GELU_PSP module are added to our network at the same time, the resulting change maps perform best subjectively, with clearer edges and fewer missed and false detections on the test target.

*2) WHU Dataset:* The ablation experimental results of the proposed method on the WHU dataset are shown in Table IV. As can be seen from Table IV, based on the SemiRCRNet, when the GELU_PSP module or the SE attention module is added, the OA value increases slightly, and the IOU value increases from 0.72% to 2.85%, but the F1 value increase obviously from 2.64% to 6.31%, at the sampling rate of 5%, 10%, 20%, or 40%.

To see these improvements more visually, Fig. 8 selects three representative graphs with a sampling rate of 40% on the WHU dataset. As can be seen from Fig. 8, when using the SemiRCRNet method, there are many missed detections and false detections in the results. This is because there are many objects with similar colors to the buildings in the WHU dataset, such as roads.

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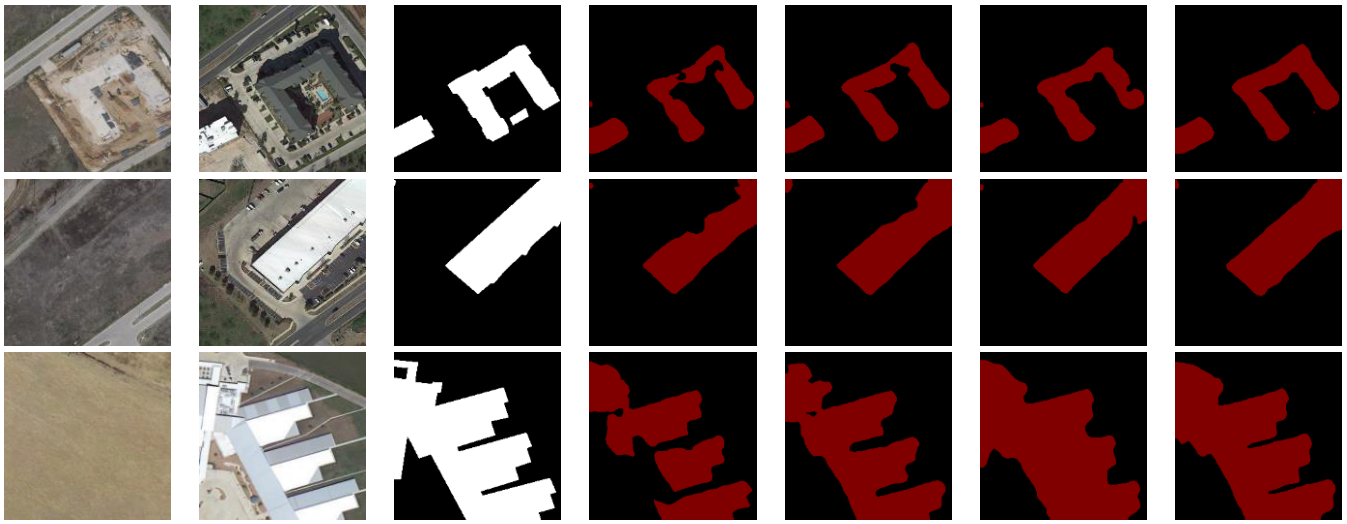



Fig. 7. Comparison of visual results of ablation experiments with a sampling rate of 40% on LEVIR-CD. (a) Image T1. (b) Image T2. (c) Ground truth. (d) SemiRCRNet. (e) SemiPSENet (SE). (f) SemiPSENet (GELU_PSP). (g) SemiPSENet (GELU_PSP+SE).

TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON THE WHU DATASET

| Methods | Module | | Labeled Sampling Ratio | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SE | GELU_PSP | 5% | | | | 10% | | | | 20% | | | | 40% | | | |
| | | | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa | IOU | OA | F1 | Kappa |
| **SemiRCRNet** | | | 0.658 | 0.9837 | 0.7294 | 0.7842 | 0.681 | 0.9847 | 0.8014 | 0.8252 | 0.748 | 0.9884 | 0.8461 | 0.8258 | 0.772 | 0.9896 | 0.8696 | 0.8335 |
| **SemiPSENet** | √ | | 0.6694 | 0.9839 | 0.7558 | 0.8115 | 0.6882 | 0.9850 | 0.8478 | 0.8301 | 0.7639 | 0.9885 | 0.8836 | 0.8791 | 0.7798 | 0.9876 | 0.9054 | 0.8864 |
| **SemiPSENet** | | √ | 0.6778 | 0.9845 | 0.7896 | 0.8207 | 0.6902 | 0.9853 | 0.8521 | 0.8265 | 0.7643 | 0.9892 | 0.8921 | 0.8898 | 0.7786 | 0.9885 | 0.9138 | 0.9058 |
| **SemiPSENet** | √ | √ | **0.6788** | **0.9856** | **0.7925** | **0.8225** | **0.7057** | **0.9860** | **0.8561** | **0.8317** | **0.7765** | **0.9897** | **0.8994** | **0.8921** | **0.7861** | **0.9905** | **0.9223** | **0.9175** |

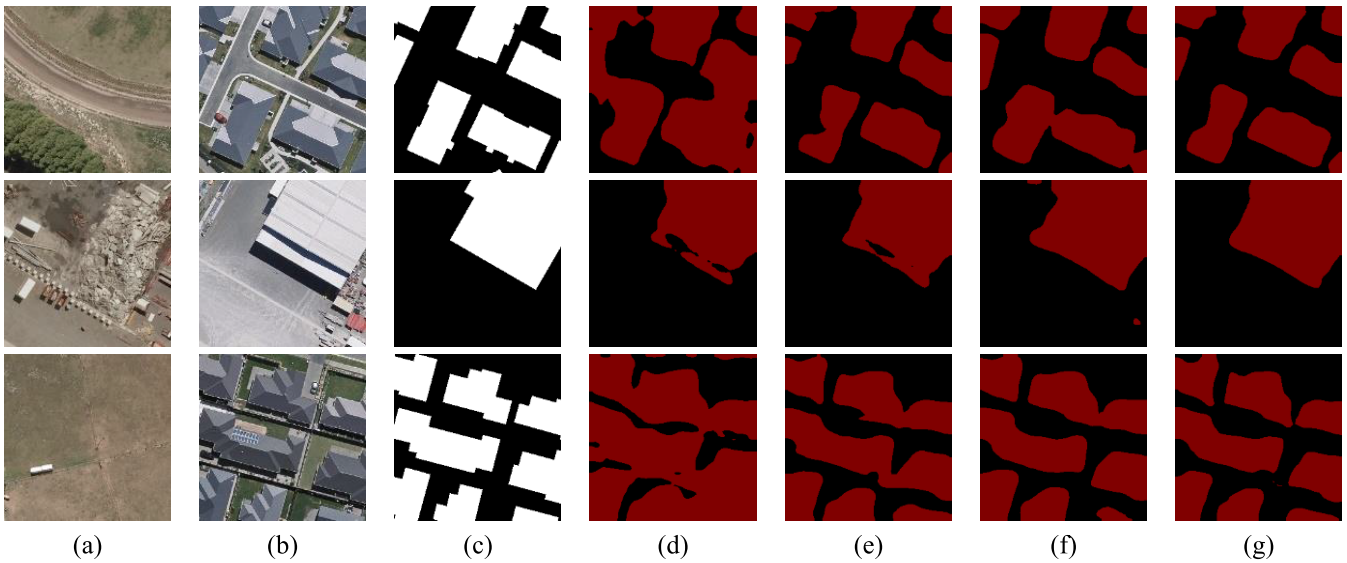



Fig. 8. Comparison of visual results of ablation experiments with a sampling rate of 40% on WHU. (a) Image T1. (b) Image T2. (c) Ground truth. (d) SemiRCRNet. (e) SemiPSENet (SE). (f) SemiPSENet (GELU_PSP). (g) SemiPSENet (GELU_PSP+SE).

Based on the ablation experiments on the LEVIR-CD dataset and WHU dataset, our SemiPSENet method can effectively eliminate the impact of some unfavorable factors. When the SE module is added between the encoder and decoder of the SemiRCRNet network, some pseudo-changes are filtered out to a certain extent. And the GELU_PSP module integrates contextual information to the SemiRCRNet network to extract more changing semantic information, which can reduce the occurrence of missing detections. And these two modules are beneficial for improving our

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SemiPSENet performance, but there is a very small increase in cost.

## V. Conclusion

In this article, we propose a novel semi-supervised network for change detection on remote sensing images, named SemiPSENet. In the SemiPSENet network, we use the GELU_PSP module for generating multiscale feature maps to integrate more contextual information, which enables the model to reduce the loss of semantic information about changes. To highlight change regions and ignore pseudo-change regions, we use the SE attention module for highlighting important change features. we use a consistent regularization method, which can make the prediction change probability graph of the semi-supervised change detection network have a consistent ability under different random perturbations, thus improving the performance of change detection. Experimental results on LEVIR-CD and WHU datasets show that the SemiPSENet can also significantly improve the performance of the network using only a small amount of labeled data and a large amount of unlabeled data. And we can summarize it as follows.

1) Increasing the number of labeled data samples in the training set can improve the change detection performance of all methods, indicating that when more labeled data can be obtained, the model can achieve better detection results.
2) Some semi-supervised change detection methods provide even better detection results than fully supervised change detection methods when the same amount of labeled data is used, and experimental results confirm that change detection can benefit from freely available unlabeled remote sensing data.
3) Our semi-supervised change detection method has advantages over the SOTA SemiRCRNet and SemiSANet methods, and experiments have demonstrated that our method can utilize information from unlabeled data more effectively. More importantly, when the amount of labeled data is very small (5% and 10% cases), our method significantly outperforms the SOTA method in terms of F1 and Kappa metrics.

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